



A split fast-convolution filter bank for the TRX-duo

Wideband analysis on a Zynq-7010 FPGA · channel synthesis on the host PC

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What fcfb is — and why

One wideband analysis filter bank runs on the board; the host reconstructs individual receivers from it. The expensive full-rate DSP happens once, in the FPGA; the flexible per-channel work runs on a commodity PC.



One capture, many receivers

A single wideband bank on the board; the host pulls as many channels as the link budget allows.



Channels defined on the host

Centre frequency, bandwidth, Kaiser filters and demodulation — all chosen at runtime.



Flexibility in software

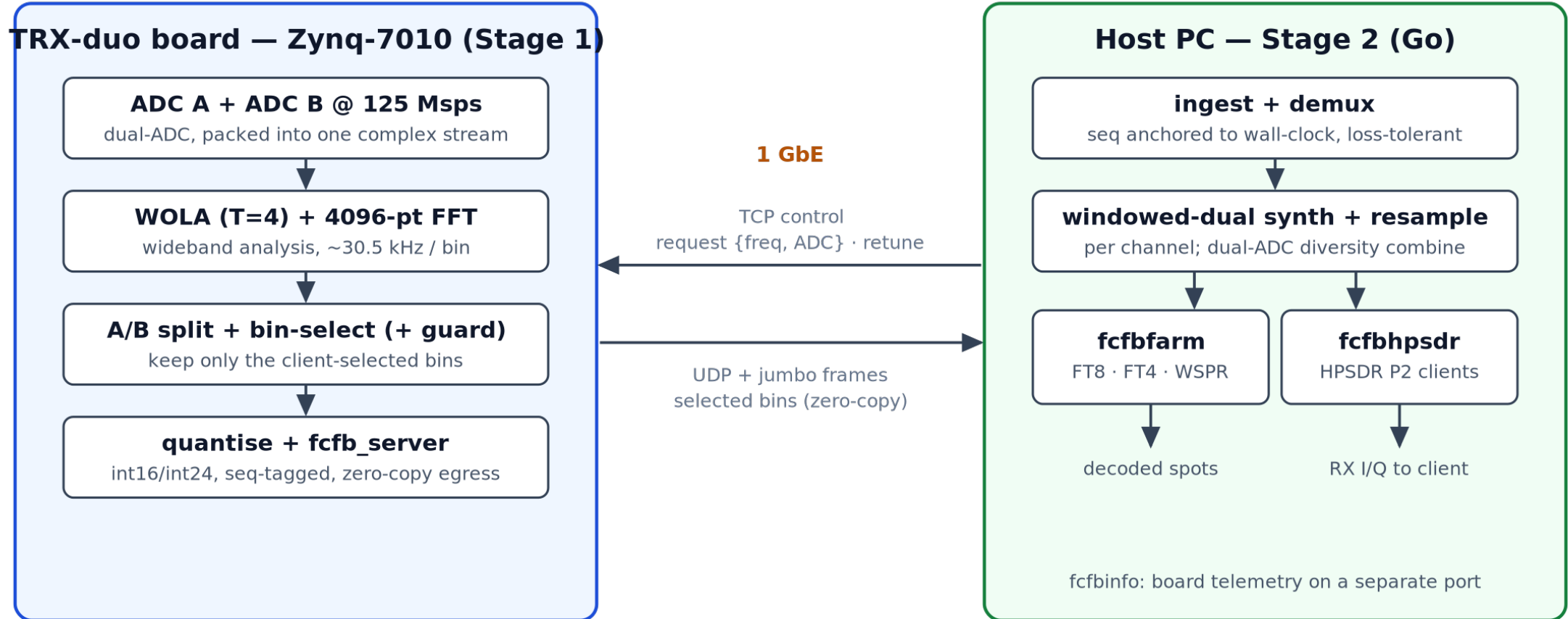
New modes and receivers are host code, not an FPGA gateway rebuild.



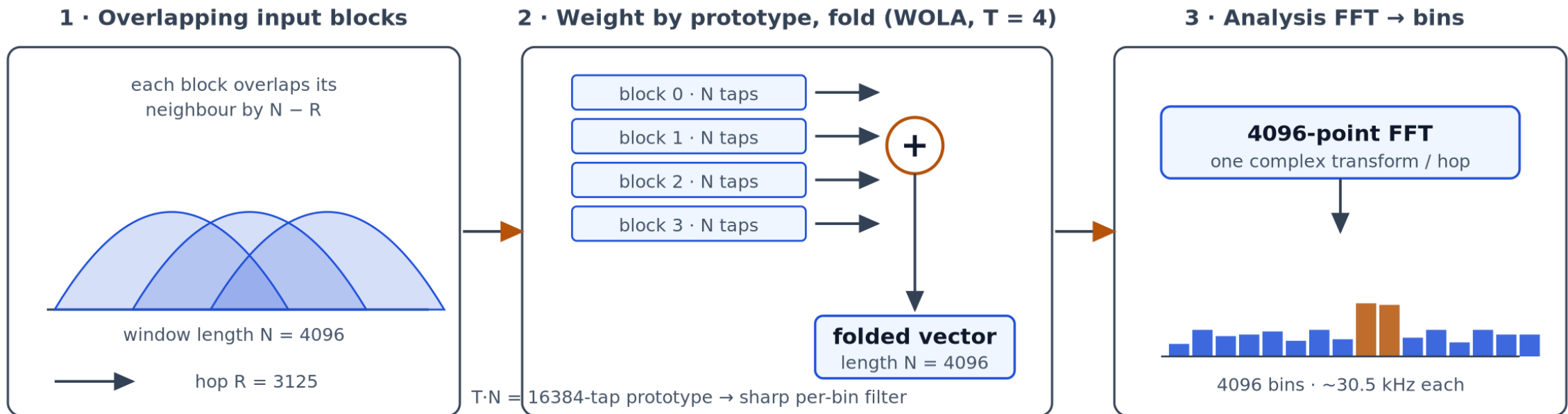
Modest hardware

An inexpensive Zynq-7010 board paired with an ordinary host PC.

Two systems: board + host



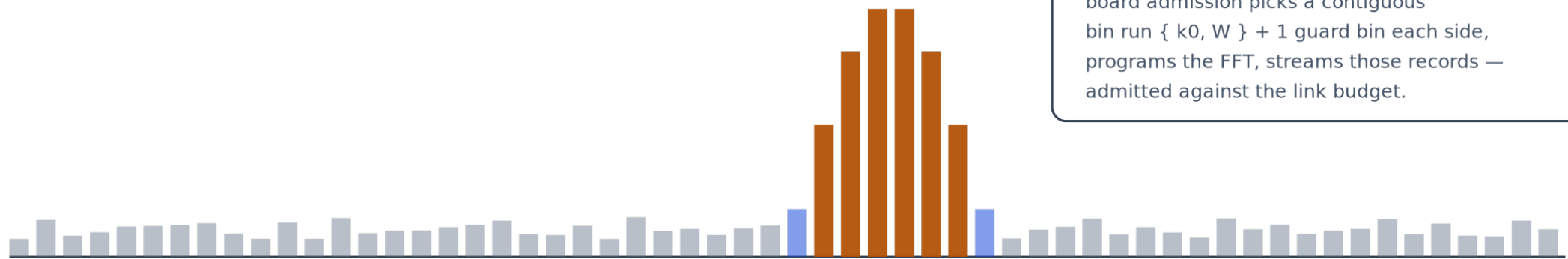
The analysis engine: overlap-save + WOLA



Fast-convolution analysis: the window slides by hop R each block (overlap-save); every FFT bin is a sharp band-pass channel the host can reconstruct.

Ship only the bins the host needs

4096 FFT bins analysed on the board — full 62.5 MHz span



noise floor — bins not requested are never sent

selected run: W bins + 1 guard each side



UDP data stream — only the selected bins cross the 1 GbE link

client requests { freq range, ADC(s) }

board admission picks a contiguous bin run $\{ k_0, W \}$ + 1 guard bin each side, programs the FFT, streams those records — admitted against the link budget.

Constraints we had to overcome



1 GbE throughput

The link cannot carry the full-rate dual-ADC spectrum. The real wall is per-datagram packet rate, not line rate: at a 1500-byte MTU a record must fit one unfragmented datagram (≤ 242 int24 bins) — fragment it and loss sets in.



Zynq-7010 resources

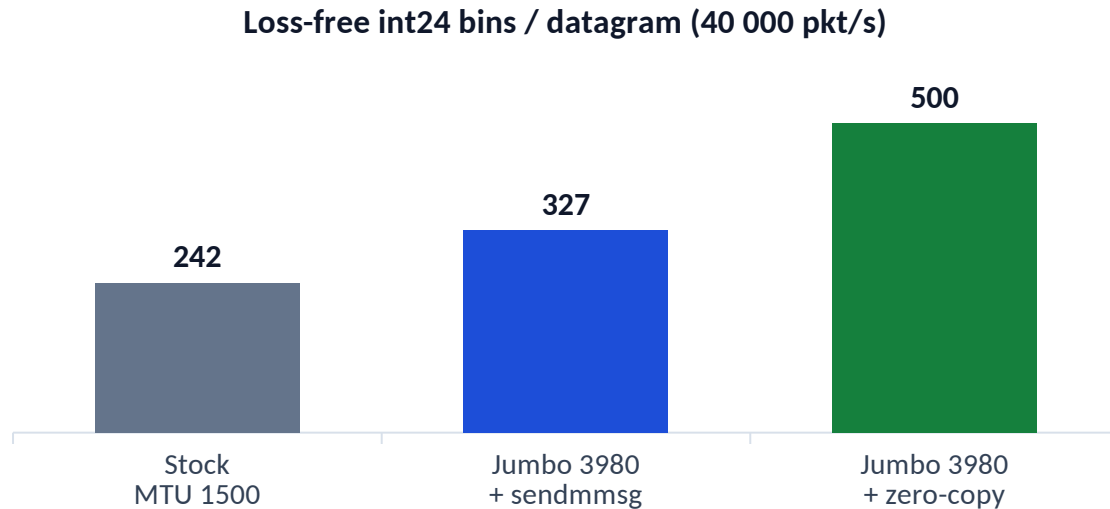
A small, inexpensive part. The WOLA prefilter is folded to its storage floor ($NB = T+1$) — the largest analysis bank that fits the 7010's block RAM.



Clock constraints

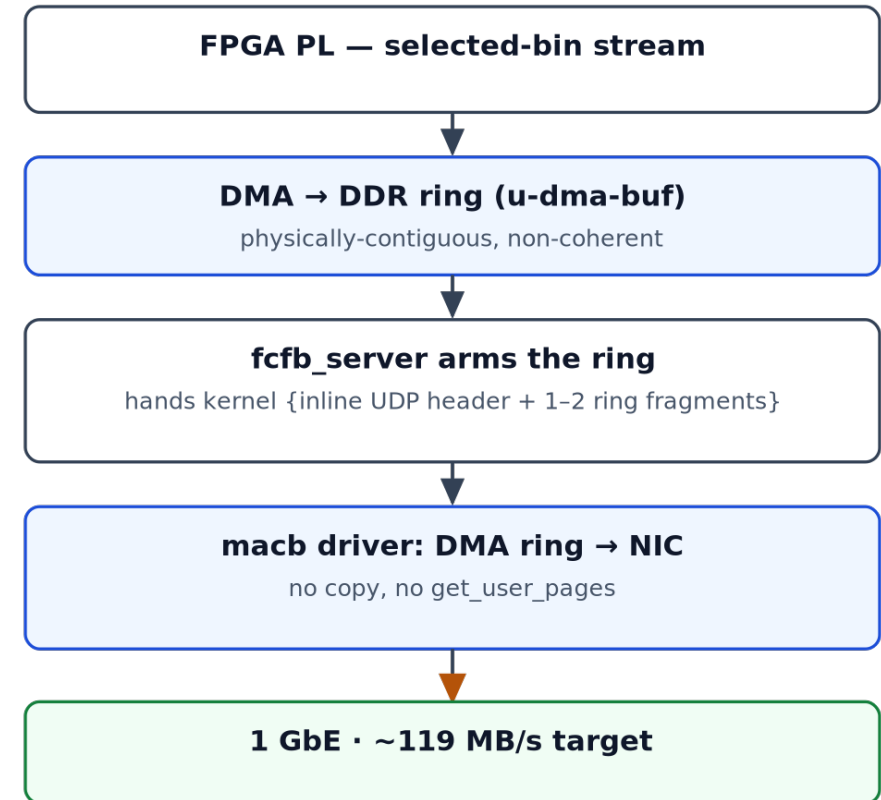
The analysis FFT runs at the full 125 Msps ADC rate: a 125 MHz datapath with 2× and 3× clock domains driving the complex-multiply FFT in step.

Network: jumbo frames + zero-copy UDP



Zero-copy moves the ceiling from the packet-rate wall to raw GbE line rate: $W \approx 500$ bins on one ADC (~ 506 theoretical, $w_{\max} 512$ just above) streams loss-free at $\sim 99\%$ of 1 GbE — ≈ 15.3 MHz. Both ADCs: 250 bins each.

Why 24-bit bins: the wideband bank carries the whole band's dynamic range through one quantiser. 16-bit samples would hold the on-wire floor near -90 dBc and limit reconstruction; 24-bit keeps quantisation below the ~ -102 dBc fixed-point ceiling — at 3 bytes per bin.



Jumbo + zero-copy: the macb driver ships with jumbo disabled; a kernel patch enables MTU 3980. A second patch DMA's each datagram straight from the ring — no copy, no `get_user_pages` — breaking the $\sim 30k$ packets/s wall so the fixed 40k/s sustains at line rate (~ 119 MB/s).

The host applications



fcbfarm
Decoder farm

Reconstructs many channels from the one board stream and runs jt9 (FT8/FT4) or wsprd (WSPR) on UTC-aligned windows — many receivers over a single stream session.



fcfbhpsdr
openHPSDR Protocol-2 server

Presents the board to piHPSDR, Thetis or linhpsdr as one or more radios; the client picks each DDC's centre frequency, sample rate and ADC at runtime.

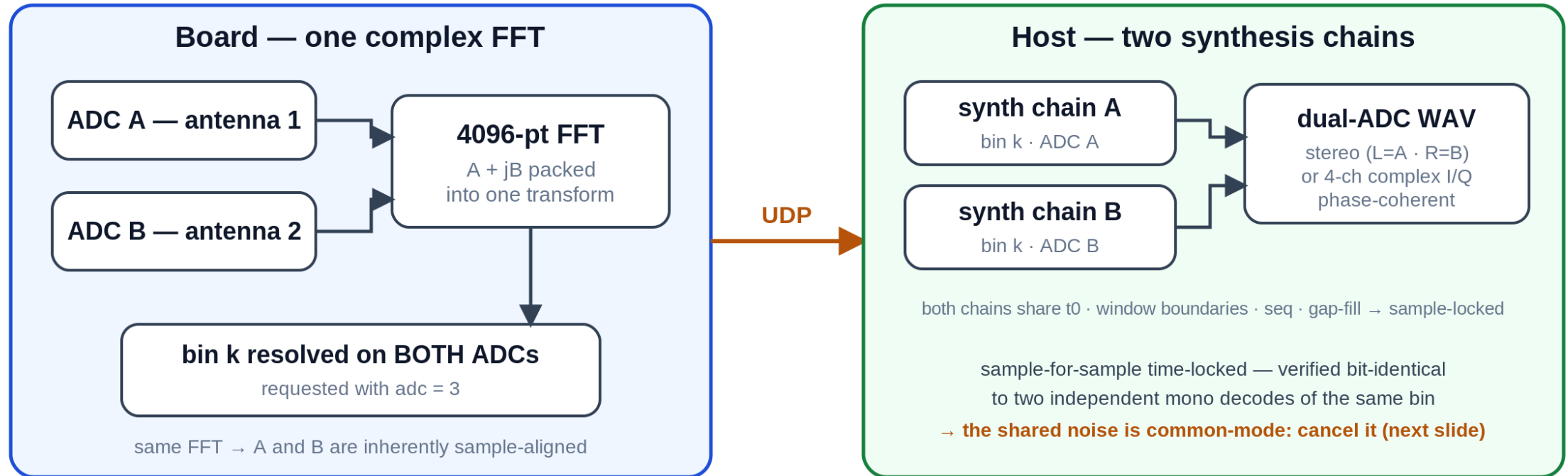


fcfbinfo
Board telemetry

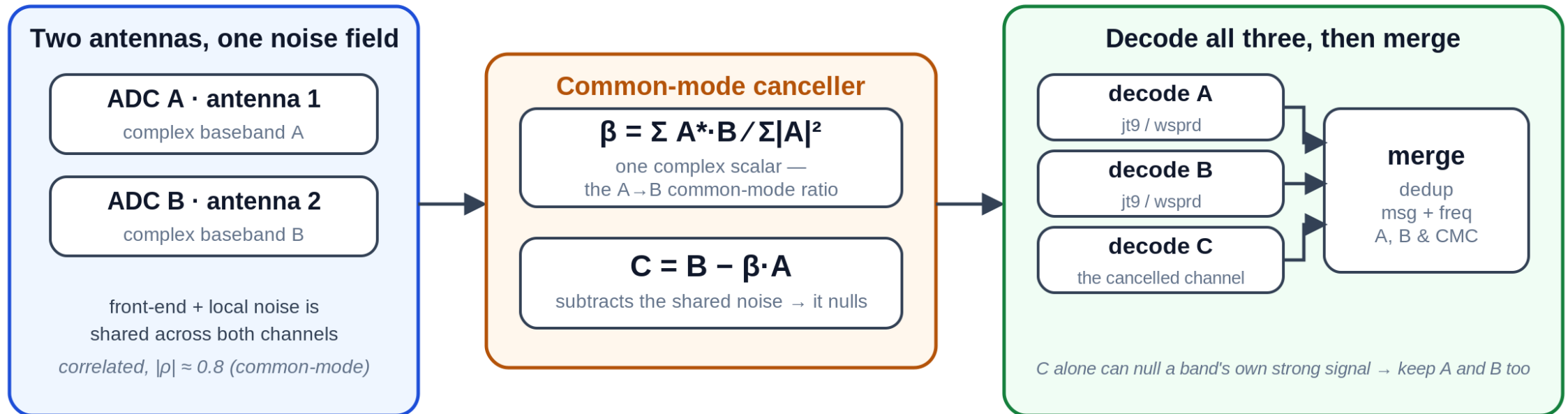
A one-shot query for die temperature, supply voltages, FPGA part and loaded gateway — answered on a separate port, safe to run mid-stream.

One zero-dependency Go package → single static binaries for Linux, Windows and macOS; the synthesis kernel is embedded, so nothing else ships.

Diversity reception



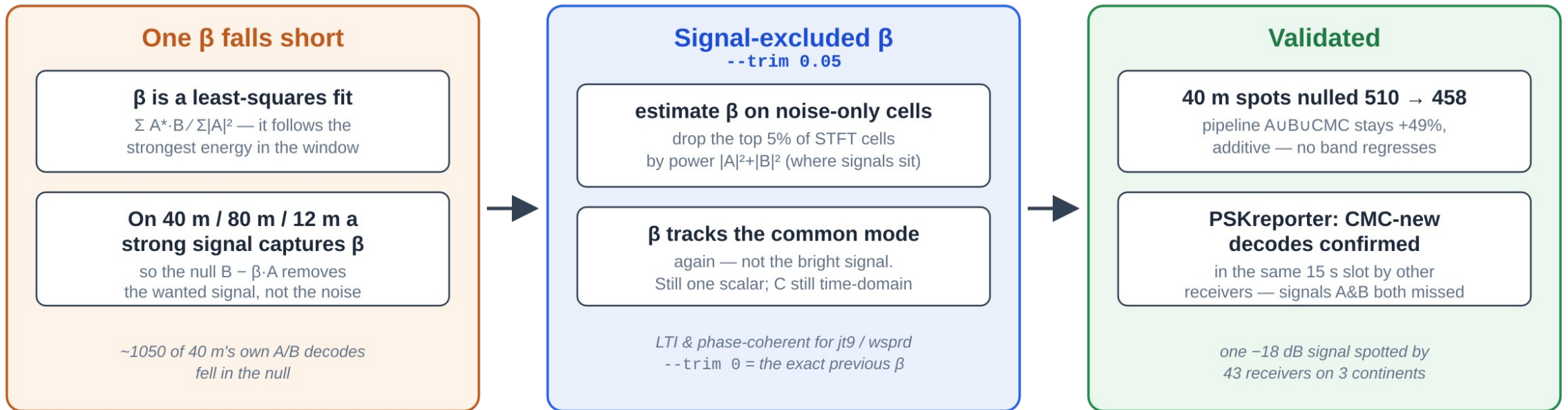
Common-mode cancellation



The null removes the noise both antennas share, uncovering weak signals buried under it

+47% more decodes than A and B alone — additive on every band

Signal-excluded β — recovering weak signals



Estimate β where the signals aren't — recover the weak ones the single null was removing

β on the noise, not the signal — on by default (`--trim 0.05`)